

Aakash Kumar Nain

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Publications

- **Transformer Layers as Painters (AAAI 2025)**. Studied representational structure in frozen pretrained transformers and showed functional differences across layers. Demonstrated robustness under layer skipping, reordering, and parallel execution, providing insights into efficient inference and transformer information flow.
- **The Ungrounded Alignment Problem (ICDL 2025)**. Introduced a formulation of alignment without explicit grounding and showed that learning systems can encode predefined behaviors using statistical structure alone. Demonstrated modality agnostic alignment through unsupervised recovery of symbolic associations.

Experience

Emergence.ai

Jan 2024 – Aug 2025

Senior Machine Learning Engineer (Research)

Delhi, India

- Worked with the **Scratch** team on the image moderation model to filter out NSFW images based on the image's content, including visuals and text. The system processes 60M images/month.
- Developed synthetic data generation pipelines and helped build the internal infrastructure for training LLMs.

H2O.ai

April 2021 – Oct 2023

Senior Data Scientist

Delhi, India

- **KYC**: Worked with the CBA team and helped them leverage **Document.ai** for extracting information from KYC acceptable documents in real-time using **Intelligent Document Processing**. This automation helped CBA process 6,000 pages per hour or 0.5M pages per week, reducing existing operational costs by more than 65%.
- **Health Claims Inspection App**: Developed and deployed a deep learning-based solution for detecting fraud in health claims for IFFCO-Tokio. The system reduced operational costs by approximately 30% and processes more than 2,000 documents every day.

Ola Electric Mobility Pvt. Ltd.

July 2018 – March 2021

Research Engineer II

Bangalore, India

- Developed and deployed a vision based ADAS forward collision warning system across more than 500 vehicles within a \$200 hardware budget.

Paralleldots (Artzen Software Labs Pvt. Ltd.)

Jan 2017 – Sep 2017

Data Scientist

Gurgaon, India

- Co-developed the **Cornea** app, an end-to-end AI-powered photo editor created for Instagram influencers. Featured in [YourStory](#) and [Product Hunt](#) with 25,000+ downloads in the first week.

Open Source & Research Projects

nanoGPTJAX

- Scalable JAX framework for efficient LLM pretraining and post training without high level abstractions. Implements modern training components including RoPE, GQA, KV cache, Muon optimizer, and CWD.

Equinox

- Contributed the Mistral-7B code example. Fixed [bugs](#) in normalization layers that caused instabilities when training LLMs in Equinox.

Keras 3.0 (Core Collaborator) / Keras 2.0 (Contributor)

- One of the first core collaborators in the Keras core [team](#) that implemented the multi-backend Keras. Contributed JAX NN ops, merging layers, metrics, loss functions, data adapters, applications, and code examples.
- Contributed **official code examples**: [OCR for captcha images](#), [Handwriting recognition](#), [Image captioning](#), [Denoising Diffusion Models](#), [WGAN-GP](#), and [CycleGAN](#).

TensorFlow (Contributor and Maintainer)

- Maintained and contributed to the TensorFlow-addons package for four years. Contributions include [Focal loss](#), [Cohen's Kappa](#), and [Gelu](#) activation layer.

Technical Skills

- **Core Expertise**: Large language models, multimodality, computer vision, synthetic data generation
- **Frameworks**: JAX, PyTorch, TensorFlow, Keras, Equinox, OpenCV, XGBoost
- **Languages**: Python (expert), C/C++ (working knowledge)

Education

GGs Indraprastha University

2012 – 2016

Bachelor of Technology in Computer Science

Delhi, India